



Department of Physical Sciences
Faculty of Applied Sciences
Rajarata University of Sri Lanka, Mihintale
B.Sc. in Applied Sciences

MAT 2306 – Operational Research I
Tutorial No: 2

1. Solve the following problems using the simplex method

I. $\text{Max } Z = 3x_1 + 5x_2$

Subject to:

$$x_1 \leq 4$$

$$2x_2 \leq 12$$

$$3x_1 + 2x_2 \leq 18$$

$$x_1, x_2 \geq 0$$

II. $\text{Max } Z = 5x_1 + 3x_2$

Subject to:

$$3x_1 + 5x_2 \leq 15$$

$$5x_1 + 2x_2 \leq 10$$

$$x_1, x_2 \geq 0$$

2. Solve the following problems using the simplex method

III. $\text{Min } Z = -x_1 - 3x_2 + 2x_3$

Subject to:

$$3x_1 - x_2 + 3x_3 \leq 7$$

$$-2x_1 + 4x_2 \leq 12$$

$$-4x_1 + 3x_2 + 8x_3 \leq 7$$

$$x_1, x_2, x_3 \geq 0$$

3. A company makes two kinds of leather belts, belt A and belt B. Belt A is a high quality belt and belt B is of lower quality. The respective profits are Rs 4 and Rs 3 per belt. The production of each of type A requires twice as much time as a belt of type B, and if all belts were of type B, the company could make 1,000 belts per day. The supply of leather is sufficient for only 800 belts per day (both A and B combined). Belt A requires a fancy buckle and only 400 of these are available per day. There are only 700 buckles a day available for belt B.

What should be the daily production of each type of belt? Formulate this problem as an LP model and solve it using the simplex method

4. Solve the following problems using the simplex method

$$\text{Min } Z = -6x_1 - 10x_2 - 4x_3$$

Subject to:

$$x_1 + x_2 + x_3 \leq 1000$$

$$x_1 + x_2 \leq 500$$

$$x_1 + 2x_2 \leq 700$$

$$x_1, x_2, x_3 \geq 0$$

5. Use the Big -M method to solve the following problem

$$\text{Min } Z = 3x_1 + 2.5x_2$$

Subject to:

$$2x_1 + 4x_2 \geq 40$$

$$3x_1 + 2x_2 \geq 50$$

$$x_1, x_2 \geq 0$$

6. ABC Printing Company is facing a tight financial squeeze and is attempting to cut costs wherever possible. At present it has only one printing contract, and luckily, the book is selling well in both the hardcover and the paperback editions. It has just received a request to print more copies of this book in either the hardcover or the paperback form. The printing cost for the hardcover books is Rs 600 per 100 books while that for paperback is only Rs 500 per 100. Although the company is attempting to economize, it does not wish to lay off any employee. Therefore, it feels obliged to run its two printing presses – I and II, at least 80 and 60 hours per week, respectively. Press I can produce 100 hardcover books in 2 hours or 100 paperback books in 1 hour. Press II can produce 100 hardcover books in 1 hour or 100 paperbacks books in 2 hours. Determine how many books of each type should be printed in order to minimize costs.

7. Use the Big M method to solve the following LP problem

$$\text{Max } Z = x_1 + 2x_2 + 3x_3 - x_4$$

Subject to:

$$x_1 + 2x_2 + 3x_3 = 15$$

$$2x_1 + x_2 + 5x_3 = 20$$

$$x_1 + 2x_2 + x_3 + x_4 = 10$$

$$x_1, x_2, x_3, x_4 \geq 0$$

(Please answer the all the question and submit the answer script to the tutorial room on or before 18th of November 2024)